ActInf Textbook Group ~ Cohort 1 ~ Meeting 24 (Chapter 10, part 2)

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Hello, greetings. It's November 11th, 2022. We're in Cohort 1 of the textbook group,
and we're coming back to Chapter 10, looking at the sections of 10,
asking any questions about 10, thinking about any overview points.
Before next week, when it will be our final meeting, and we'll talk about feedback,
fill out the form in future textbook groups, talk about project ideas, next steps,
talk about our plans for 2023, and so on.
But where do we begin with 10?
We covered many of the sections once last week.
There's a lot of sections.
The chapter kind of is like a symphonic presentation.
It has multiple movements and subsections that have their own rhythm and structure.
Let's go to the end.

Let's go to the end.
I think this is a really important note about learning and active engagements,
which is if you're not, even before skin in the game and psychological ownership and so on,
if one is not making predictions,
not just about material, like what's going to happen next in the TV show that I'm watching,
oh, it surprised me,
but about the consequences of one's actions in the world,
they will not be able to update those models of action in the world.
So you could be surprised and end up being a good inferor in a passive inference setting.
However, doing is what actually allows you to learn in a totally different way,
because whether you realize it at the beginning of the journey or not,
or whether you even agree with this or not,
those actions that you take are chosen and exist on a trade-off frontier
of epistemic and pragmatic value.
And so simply choosing to act
is a strategy
that opens up a space
for pragmatic as well as epistemic action.
It's kind of like the hyper-prior-on-active inference is how active.
And one other note,
just that anyone can add anything.
This was a discussion with JF and Candon

some days ago in our research meeting
about the scaling debate in machine learning.
All you need is scaling.
Bigger natural language models and so on.
And like,
how can active inference
and how can we
contextualize that
and improve the state of things?
And
we talked about how,
importantly,
GPT
does not choose its own inputs.
And so,
yes,
it can engage within a perception action loop
related to receiving inputs and sending outputs.
However,
in terms of its own training and development,
it's importantly
constrained.
And that's not necessarily a bad thing.
It's just that
to actively select
what stimuli to sample
is very central to active inference.
From the ocular motor
to the tactile.
And
so,
all of the big data training algorithms
do not select their own inputs.
And then if they were to,
it could be done in two ways.
It could be done in an ad hoc way
or principled way.
The ad hoc way would be

you do some kind of
different architecture of a model
that,
you know,
predicts what inputs would be valuable to train.
or
you could think of a unified imperative
that would decide
on both action and learning rules.
Eric and Brock.
Yeah,
I suppose you could say that reinforcement learning,
the machine is selecting its own
samples to get feedback on.
Now,
reinforcement learning is not GPT-3.
It's not language learning,
language modeling.
but
I think an earlier paradigm,
maybe I don't know.
I think they still use reinforcement learning
and deep learning models.
Although,
you know,
it was also,
I think a lot of people are discouraged by that
because it's,
it,
you know,
as they say,
it's a,
it's a very narrow,
a very low bandwidth signal you're getting back.
But it is an example of learning models using,
generating their own samples.
Thanks,
Brock.

Um,
I was gonna,
I guess,
uh,
take it to this biological place we left off in last session
of the circadian of like,
being a kind of data cleaning process almost,
or,
um,
stabilizing the,
you know,
the like data set coming in.
Um,
and,
um,
that we do have some
very,
very crude,
like,
uh,
you know,
built-in labels
for edges
kind of in our visual cortex,
right?
Uh,
that are
presumably
evolutionary.
Um,
and some decent evidence that
like maybe one level higher,
something gestalt-like of like,
edges and heights,
you know,
um,
or,
uh,

yeah,
just certain kinds of like,
really,
really,
uh,
generalized,
um,
images or views that you might take
that have like,
very pronounced,
uh,
definition,
edges of like,
like a,
a prayer,
you know,
like a landscape of,
you know,
our natural evolutionary landscape being like somewhat comforting or pleasing or something like this,
right?
And walking over like a baby,
like eight months or whatever,
crawling over a glass,
but it looks like,
looks like they're gonna fall
and they're,
they react to that,
right?
Um,
like these sort of things,
uh,
where there's some sort of baked in,
like,
priors or,
again,
like labeling,
right?
Um,

there's no,

I mean,

are those hyperparameters,

you know,

there's,

but then they're not controlling their hyperparameters

in these models,

like,

so,

um,

yeah,

I mean,

there's no affordance there for that sort of like goal directed or selecting data sets in any possible way or any of that.

So.

I'm thinking of a,

of a British English reading of how the world should be.

Like,

we talk about expectations and preferences.

And if someone says,

like,

I should think so,

it's not actually a normative

stance.

It's like saying,

I,

I expect and prefer myself.

I should be there at eight.

Whereas often should is interpreted as purely normative.

like,

I wish that you did this.

Um,

but should

actually does kind of condense that expectation preference,

um,

scenario.

And then to kind of come back to the,

um,

inherited world models.

So,

yeah,

um,

sequential processing steps can be understood of,

within a hierarchical processing framework where higher levels have the interpretation as being hyper parameters on lower levels.

And I think we saw a little bit at the end of,

um,

nine,

which is basically for any,

and this also speaks to the composability of Bayes graphs,

which is that you can always have a variable as fixed.

or you can have a hyper prior on that and an uncertainty on the hyper prior,

or you could wrap it in another Bayes graph,

but all of these can fall under the category of how things should be for that agents.

And a business that thinks that there should be,

um,

um,

a lot of interest next year and is wrong goes out of business.

the organism that thinks that it should be warm tomorrow night and doesn't plan for the cold dies.

And so the,

the should almost reflects the whole cognitive stack.

And the action selection that makes it that way.

one area that,

yeah,

I'll leave,

please.

uh,

yeah,

uh,

yeah,

I was,

uh,

reading the other day about,

uh,

the importance of,
uh,
the statistical,
the regularity of,
uh,
any particular environment and,
uh,
the way it constrains,
uh,
the evolutionary developments.
Uh,
I mean,
uh,
for instance,
for terrestrial animals,
uh,
uh,
the majority of the objects are solid objects.
And,
uh,
of course,
that's not the case for the sea animals,
but,
um,
uh,
this,
uh,
I was wondering that,
uh,
when an organism generates it or constructs its,
uh,
its generative model,
uh,
how exactly,
especially in the case of visual perception,
uh,
how exactly these kinds of statistical regularities,
uh,

or,

well,

let's say,

uh,

the,

uh,

statistical regularities as applied to their visual perception,

uh,

translates into their generative models,

uh,

because,

uh,

uh,

for the most part,

uh,

at least in this textbook,

what we saw,

uh,

in generative model is just,

uh,

exactly,

uh,

it is just,

uh,

uh,

an abstract,

uh,

conception of generative model.

And,

uh,

as you say,

how the,

how the world should be,

uh,

uh,

but,

uh,

uh,

I don't see how these kinds of,
um,
um,
evolutionary,
uh,
constraints,
or statistical,
uh,
regularities of the environment,
uh,
takes place,
uh,
uh,
when a generative model,
uh,
is constructed at each,
uh,
for each specific task,
uh,
because,
uh,
in my view,
that would make a lot of difference in,
uh,
constructing the other priors,
according to those regularities.
Uh,
I'm not sure if I express my question clearly enough,
but,
uh,
yeah,
I was wondering how that's done in active inference framework.
Yeah.
Good question.
Um,
just on the visual note and then the generative model on the visual note,
there are some statistical regularities of natural visual scenes,
like pebbles on the ground,

trees,

like there's some auto correlation functions and some spectral density analyses and some fractal similarity things that people have done.

So that like for quote natural scenes,

so not necessarily a skyscraper,

but again,

like the bodies of animals and,

um,

natural settings with trees and plants,

there are some like hyper parameters that make the scene natural,

which is interesting to think about in these like image generating algorithms.

And then,

um,

to the second point about the generative model,

um,

it reminds me of JF's approach towards society,

where it's like spinning up generative models.

Now in that case,

it's symbolic,

logical generative models,

but also we could imagine society of statistical generative,

generative models.

Um,

Eric.

Yeah,

I would just,

um,

echo,

um,

Ali's question about how,

um,

um,

or the,

you know,

a,

a,

a,

a gap in,
you know,
what we've,
we've learned about active inference,
which is how do these models get made in the first place?
Um,
you know,
for the examples in the book,
um,
well,
he walked in with the team A's and a model for how the team A's works.
And that was probably the most sophisticated active inference model presented.
Um,
but when you get to,
you know,
real world scenarios,
well,
how,
how are models of the world learned?
And that's a big open question.
And,
you know,
are they even,
what sense are the generative models?

often predictive,
I think,
but,
um,
um,
you know,
are they probabilistic generative models in the sense that we think of them mathematically?
I,
I,
that's,
you know,
some model people model it that way.
Some,

some people don't,
don't think so.
Um,
you know,
certainly neural networks are discriminative in many cases.
Um,
so I just,
I guess agreeing that that's,
there's a lot of,
a lot of work unsaid or,
um,
to,
to be done there or a lot of questions open.
Hmm.
I wonder if some people hope for perceptual control theory to kind of save the day,
which is like,
we don't need to make a generative models for the bewildering possibilities of external states,
but it might be enough first to have bodies that can engage in interoception and homeostation.
And then,
and then,
with that machinery as,
as enabling,
maybe all you need to know about the external world is like,
it's good or it's bad.
And just like run away from bad things and go towards good things.
And then you have to,
you know,
you can update your A matrix on whether you think good or bad states emit this or that kind of observation.
but,
um,
the,
the blanket
facilitates
the organism's hyper focus on its own states
and might allow it to engage less in this open-ended structure model fitting
of the world.
not that you wouldn't do better with infinite resources and a better causal model of the world.
Brock?

Um,
what's something that we kind of discussed with JF actually too,
that,
um,
some of his generative models are going to be static and specified,
just like the robot is literally specified by him.
Um,
it seems like if you wanted to,
you know,
have a more general explanation or process or whatever to get to whatever human level or something,
things that start exhibiting those sort of behaviors or,
or,
uh,
uh,
you know,
commensurate with active inference type models of like,
you would need evolution.
Like,
like you would need to evolve the system to that point.
And then
you could ask a question like,
if you did that,
would you gain anything
compared to
just build optimus
and it's not as good,
but it's a good,
like first order approximation,
first step
of like,
towards something that
would be constrained in such a way
that it's more likely
to produce generative models that are useful and
whatever,
could be human-like or,
or whatever,
you know,

um,
yeah,
conscious,
intelligent,
something,
whatever.
um,
but
I don't see how,
yeah,
I mean,
any possible explanations
on the table for like,
how you get to
something that's
whatever,
that has some sort of central nervous system,
you know,
or,
or conscious or anything like that,
um,
that would all depend on something like evolution.
And,
and then that process is just necessarily like very long and laborious.
And if you're just,
if you're looking for something novel,
then maybe you have to do that.
But if you're just trying to get the first order approximations,
something similar to what already exists,
then why go through all that?
You know,
it's probably easier to just assume some priors,
which is what JF is doing.
So this reminds me of the problem in machine learning of one shot learning,
where like a single image is shown.
And the goal is to learn cat as a category from one picture of a cat.
Now there are algorithms that do better or worse,
but there's also some fundamental limits of any type of one shot learning.

Like to say nothing of overgeneralizing,
you know,
overfitting,
but of course.
So it's almost like this is,
um,
statistics slowly retreating or advancing into biology because it's like,
well,
if,
if we don't want to do just one shot learning on one data point or one data
set,
which is one datum still,
it's one big datum.
We want to have a learning model.
Okay.
Now we want to have a learning model with active selection of inputs.
Now we want to have,
um,
generators and basically like landscapes of models,
portfolios of models,
Bayesian model reduction,
and structure learning in the Bayesian setting as being across distributions of models.
And then that's where evolution gets us,
which is populations of models existing in different niches.
And maybe if our data set is our generative process,
like maybe there are some models that are doing like really well in this part of the data set.
And there's other models that are doing really well in another part of the data set.
You don't just compromise them or,
um,
you know,
ligate them together.
Their embodiment reflects the statistical regularities of the environment.
And I added this,
um,
citation.
This is from professor Gordon,
my PhD advisor.
And it's like,

and there were several subsequent papers that continued on this theme,

but it was like trying to get without putting words into her mouth or anything like that,

like trying to make this point that you couldn't,

you can't just look at the collective behavior or the end of,

you know,

everything's collectives all the way down.

So you can't just look at the behavior of an ants and just say like,

whether it's like good or bad or fit or not,

because there's landscapes with uniform and patchy distributions,

which is exactly the case for statistical distributions.

There's cases where the cost of stopping is high or low relative to the cost of operating.

And then there's situations where basically interruptions can be easily recreated or not among others.

Um,

and it just like,

the generative process is what the generative model is fit to.

So it's an incomplete sentence or an incomplete expression to be like speculating on just generative model architectures.

Without a little bit of,

at least an acknowledgement of the statistical regularities of the generative process.

If the generative process is a metronome that's alternating between states,

then your generative model,

you have some suggestions for how to design it.

If the generative process is another agent like you,

then that's another type.

That's like the generative adversarial neural network case or the game theory case or the theory of mind case or the communications case.

Like those are different settings.

Um,

returning to chapter six,

what are we modeling?

What is the appropriate form for the generative model?

How to set it up and then how to set up the generative process.

And I think we even discussed a little bit then,

like,

this is kind of a generative model first approach.

Whereas one could imagine step four moving to step one and a half.

Or,

you know,

however,

but,

but,

um,

it doesn't make sense.

You know,

nothing in biology makes sense except in the light of evolution.

Famous Dobzhansky quote.

So,

will we have similar memes?

You know,

nothing about the generative model is going to make sense except in light of a generative process.

the state transitions in the generative model,

they are parameterized given the weight of the leg.

That's not part of the parameterization of the state space.

But in fact,

that parameterization makes sense in light of gravity,

friction,

the solidity of objects.

And so the,

the disembodied general,

generalized,

uh,

general generative models,

we lose sight of it because it's like,

it feels like we can totally define the whole setting.

Therefore,

the parameters of the generative model should be like maximally interpretive.

But that might not even be true for the in silico case.

So especially when we're taking the metabasean perspective from chapter nine,

and we're thinking about doing inference on an inference doer,

there's a lot of,

you know,

X only makes sense of Y,

which only makes sense in light of Z.

And I think it's actually that more complex predicate.

I'm not sure if that's the accurate way to say it,

but it's the more complex setup in the metabasean model.

That's one of the energy barriers to understanding,

not just act inf,

but just good science.

And the map territory.

And all these other related topics.

Especially because this dashed line,

that's our own gray zone.

So our own parameterization is only going to make sense in light of things that we almost surely didn't specify.

Even our own parameterization about it ourselves.

And so like understanding,

I mean,

then to kind of come all the way back,

I don't think it only leads us to an existentialist,

like who are we?

What are we?

Where are we going?

Why are we doing that?

But of course,

that's one of the places that you'll get to in this road network with a high road and the low road.

Yeah.

Yeah.

Yeah.

One area we didn't go too much into,

which is totally understandable given just that it's a textbook that we've read over the last several months and all that is like the citations.

And just,

okay,

you know,

structure learning.

Tenenbaum 2008.

Well,

we know Tenenbaum is still active.

So,

and it's always useful to cite earlier papers that are more positional.

However,

there's a lot of threads that could be connected to like,

I don't know,

a 2019 review on structure learning techniques.

so that the framing of the question and the comparators might be a little bit more modern.

Learning to learn in machine learning.

1949.

Formation of learning sets.

Learning to learn in machine learning.

the seriously brain damaged monkey developed sets,

which allowed him to behave more efficiently than the untrained non brain damaged monkey.

brain damages adaptive.

Is that what?

This is the abstract.

It's not the conclusion,

but

I just can't help it.

Let's.

Yeah.

I don't know.

Like,

I can see that the clickbait headlines already.

I don't know.

I don't know.

Continuing this,

the title of the chapter.

And,

um,
another question about categories from the last,
uh,
session here.
Like,
what?
What?
Do,
what is,
is there a specific active inference definition of sentient?
how does that relate to,
you know,
attention salient epistemic dynamics and not going down that existential lab hole we just left off on?
Uh,
I don't know.
Ollie.
Dr.
Sentient.
Ollie,
what's,
is there an active,
specific,
uh,
I think he dropped off for the second.
He's back.
If you could talk,
Ollie,
but I,
I just,
I joke because I know Ollie was,
well,
um,
the upcoming lives,
the guest stream that we'll have with the,
um,
authors.
Okay.
So that they recently had this papers in vitro neurons,

learn exhibit sentience.
when embodied in a simulated game world,
they played pong.
And,
and then like people were,
um,
yeah.
Okay.
People were freaking out.
Because it was just like sentient as conscious,
but Ollie,
what is the take?
Well,
uh,
I'm,
I'm not exactly sure what,
uh,
I mean,
what,
what was the controversy about?
Because,
uh,
it's,
it was merely about just,
uh,
a lexic,
lexical meaning,
uh,
of a term,
uh,
which is,
uh,
obviously something that,
that you can look up,
uh,
in a dictionary and get those lexical meanings.
Because,
uh,

the word sentient,
uh,
is,
uh,
defined in Oxford English dictionary as,
uh,
an organism that responds to stimuli.
That's it.
So,
one,
at least one of the,
uh,
meanings of,
uh,
word sentient.
So,
yeah,
uh,
I was not sure,
uh,
why that,
the use of this word,
uh,
raised the,
I mean,
so much controversy on Twitter and around,
because that's basically how this word was used,
even in,
uh,
19th,
uh,
and 18th century.
So,
uh,
I mean,
before all these,
uh,
developments in artificial intelligence and philosophy of mind and so on.

So,
uh,
that's just minimal requirement of,
uh,
for something to be described as sentient.
Yeah,
I,
I guess,
um,
this framing,
distinct,
sentier,
you know,
in Spanish,
me siento,
I feel a certain way.
It's situated in a,
not just a cognitivist,
but a phenomenological space.
And so,
I believe that's some of the baggage people brought,
but you're totally right.
going to another mainstream source.
It simply is the responsivity to sensations.
Is that like exhibiting agency?
I don't know.
Exhibiting policy selection,
you know?
I don't even think it has to act.
Something could be sentient and not take actions.
Although one could then critique and say,
if it doesn't take internal actions or covert actions of some kind,
then was the impression made?
Or was it just,
you know,
Teflon just sliding off of its perceptual apparatus?
So,
but,

I mean,
it is what this chapter's title hinges on.
Yeah,
it is.
I'm not sure,
I'm not sure that we,
I don't know.
Yeah,
it's,
it's still a little confusing in the light,
in the context of Act of the Vifference to me,
I think,
like,
it would have to be like the case that,
like you're saying,
you know,
some sort of action was taken,
maybe it's a hidden state or something,
you know,
from the,
but,
uh,
I don't see how,
yeah,
like,
it could be responsive to stimuli,
and not act.
Like,
um,
because,
I mean,
I don't know,
like a rock rolling down a hill is responding to the stimuli of the,
you know,
gravitational geodesic there.
It's like,
is that,
you know what I mean?

Like,
that's not really,
um,
that's not a response to stimuli,
right?
So,
that's probably outside of the definition there.
I don't know.
Maybe in Bayesian mechanics,
it's not,
but.
Does sentient even add anything?
And are we only talking about organisms?
Organisms in the Kantian definition is the unit of biological organization that is the end in and of itself.
Not that that necessarily helps.
Whereas a more modern replicator evolutionary definition describes the organism as,
like,
the evolutionary unit or replicator,
um,
like the ant colony as an organism.
Um,
and then,
Ali,
I just saw you linked agential realism.
I mean,
could this have just been,
one of the benefits of ActInf is it provides,
again,
we talked last week whether it's complete,
um,
provides a solution to the adaptive process,
problems that agents have to solve,
or inactive agents,
or active entities,
or active inferrers have to solve.
But not,
not sure whether this
brings in some lateral associations

and over-specifications that aren't necessarily entailed by simply active inference.
actually,
uh,
uh,
Karen Barad,
uh,
I mean,
uh,

the physicist to propose this,
uh,

ontological stance of agential realism,
defines agency,
um,
as a kind of relationship between objects
and not exactly as something that,
uh,
subject or an organism,
uh,
has,
or,
or capacity of an organism,
or a cognitive,
uh,
agent,
so to speak.
So,
uh,
in this sense,
uh,
if we look at,
um,
this definition of agency,
uh,
we can say that,
uh,
every kind of organism

and every kind of,
uh,
even,
uh,
um,
persistent objects in various timescales
would necessarily have some kind of agencies.
Well,
agency is another polysemous word,
of course,
um,
because it can refer to the agent-like behavior,
meaning something you'd specify more as like a mobile actor
in a agent-based model,
but also it is then used to specifically also mean taking action.
And like in the Bayesian physics,
the textbook does not engage deeply with Bayesian physics.
And it will be interesting to see in the coming one to three years
as,
um,
Dalton's,
et al's work becomes percolating through the space
and potentially as reflected by Second's textbook
with Friston and maybe others,
um,
at MIT Press,
um,
whether the,
the different specified kinds of particles
help us
step outside of some of the previous framings of agent and agency.
So if agent is just going to mean thing,
we have a way to talk about thingness
in terms of its repeated measurements,
quantum things,
thermodynamic things,
cognitive things.
So we have a good thing definition.

Friston 2019,
which is cited in this textbook,
but it's not gone into too much.
Um,
and then we could talk about mode matching,
mode tracking,
mode tracking,
and path tracking things
and chaotic things and strange things.
And so,
um,
a piece of wood floating down the river
is following a path of least action
that doesn't entail any cognitive model.
It is a thing,
it's agentic,
but it's not taking agency.
if it were,
um,
a stick
with,
um,
a different apparatus
or increasingly different kinds
of policy selection mechanisms
that wouldn't change its thingness,
but it would be a different kind of thing.
Um,
also,
I,
I,
um,
hope it's okay to share that
and anyone who's watching,
uh,
on Monday
at,
um,

6 30 AM PT,
Maxwell Ramstead
is going to be giving,
um,
a meeting
at the theoretical neurobiology group.
So email active inference
if you want more information,
but he's going to be talking about
some of these areas.
Basically,
as part of
this
banner year of 2022,
um,
I believe there are some
threads around,
Ali,
you can totally say it,
say it differently if you,
if you see it differently,
but like,
um,
kind of standing with two feet
in
Act Inf and FEP
as a framework
and a filter,
as Dean might say,
rather than
this
Procrustean
journey
of trying to
stretch and
distort
what FEP
is

into
other
ontologies
that might have
false positives,
negatives,
no quantitative
definition in sight,
and so on.
And like,
just the way that so much,
um,
ink and attention
has been spent on
framing FEP
within
past lineages
of debate
or trying
to bring it to
bear on one
side or the
other.
But I think that
reflects a time
coming when a
true FEP
interpretation
actually is
possible,
rather than
seeing FEP
as interpreted
by others.
But that will be
kind of
interesting.
Um,

okay,
well,
we have
five more
minutes.
What a
year.
It seems
like going
faster
would have
been
challenging.
One
chapter per
week would
have been
a blitz.
Three
chapters per
week,
I mean,
I mean,
three,
three weeks
per chapter.
Yeah,
exactly.
Um,
three weeks
per chapter
would have
taken us
into the,
um,
you know,
two-thirds
of a year

range.
with some
modifications and
improvements that
I hope everybody
will engage in
and,
and have so
much to
contribute to,
like in
future textbook
groups and,
and other,
um,
affordances.
This is
something like
a semester-long
course.
course.
And that,
um,
like a
four-unit
course or
something,
where for
two weeks
reading a
chapter with
a few other
selections.
So,
providing some
other work
and maybe
starting slightly

differently.
Maybe,
but these are
all things that
we can explore,
just kind of,
like,
raising them.
I know that
all of you are
thinking about
this too,
but just wanted
to,
like,
say it.
um,
within the
context of a
semester course.
And initially
more like a
seminar course,
like,
we're working
through it,
and some
people,
like,
read and
understand highly
all the
material.
Other people,
like,
um,
fight through
and get

through some
of it,
but it all
just is what
it is,
before becoming
more of,
like,
a graded
course.
Not even
saying that
we're going
to grade,
I'm just
saying,
like,
there is a
format
archetype
where challenging
material is
addressed,
usually as a
graduate seminar,
and you
basically pass
if you
participate.
And it
doesn't require
tests at all
before we get
into other
kinds of
assessments and
everything.
Um,

any thoughts
in the last
minutes here?
Uh,
thank you
for doing
this and
having the
cognitive
range to
the answer
between all
these disparate
gestures that
we throw at
you,
and,
uh,
yeah,
I really
enjoyed it.
Thank you.
And also,
just one last
night,
I maybe
said it for
some earlier,
but,
um,
we're in
contact with
the authors
and with
the publisher.
And,
um,
like,

I sent them
some updates.
They're super
excited,
um,
so we could
schedule time
to all talk,
potentially as,
like,
recorded,
but not
live-streamed,
um,
in early
23 with
at least
Thomas,
if not
others.
And,
um,
we can
return next
week to
some of
the future
steps and
things like
that.
But it's
just,
like,
it'll be
cool.
All right,
next week.
All right.

Thank you,

fellows.

Farewell.

Bye.

Thank you,

everyone.

Bye.